

PADANY, Russian Federation

WMO: 226190

Lat: 63.2668N

Lon: 33.4101E

Elev: 130

StdP: 99.77

Time Zone: 3.00 (E03)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.0	(c) -25.0	(d) -30.4	(e) 0.2	(f) -28.0	(g) -27.3	(h) 0.3	(i) -25.0	(j) 10.1	(k) -6.2	(l) 8.4	(m) -7.3	(n) 1.5	(o) 300	(p) 0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.5	25.7	18.6	24.0	17.8	22.2	16.9	20.1	24.2	18.8	22.5	17.5	21.4	2.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	13.5	22.6	17.2	12.5	21.5	15.8	11.4	20.0	58.0	24.2	53.9	22.3	49.8	21.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.4	(b) 7.9	(c) 6.8	(e) -30.8	(f) 28.6	(g) 3.5	(h) 2.4	(i) -33.3	(j) 30.3	(k) -35.3	(l) 31.7	(m) -37.3	(n) 33.0	(o) -39.9	(p) 34.8		
(4) 9.4	(5) 7.9	(6) 6.8	(7) -30.9	(8) 20.9	(9) 3.5	(10) 1.6	(11) -33.4	(12) 22.1	(13) -35.4	(14) 23.0	(15) -37.4	(16) 24.0	(17) -39.9	(18) 25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-11.0	-9.1	-4.9	1.4	8.4	13.1	16.8	15.2	10.4	3.3	-1.2	-6.2
	DBStd	10.60	7.60	7.52	5.58	4.18	4.84	3.98	3.86	3.36	3.15	3.54	4.01	6.89
	HDD10.0	3089	650	533	462	259	88	13	2	2	33	208	337	501
	HDD18.3	5606	908	767	721	509	308	163	76	106	237	466	587	759
	CDD10.0	571	0	0	0	0	40	107	212	164	46	1	0	0
	CDD18.3	47	0	0	0	0	2	6	28	11	0	0	0	0
	CDH23.3	222	0	0	0	0	18	33	129	43	0	0	0	0
	CDH26.7	33	0	0	0	0	4	4	17	8	0	0	0	0
Wind	WSAvg	3.1	3.0	3.0	3.3	3.0	3.1	2.9	2.6	2.8	3.1	3.8	3.6	3.4
Precipitation	PrecAvg	533	26	21	24	30	37	62	66	81	61	54	38	31
	PrecMax	827	50	57	60	77	88	145	184	211	147	105	75	62
	PrecMin	369	6	2	2	4	6	15	12	12	10	1	6	8
	PrecStd	96	11	11	13	17	17	31	34	42	28	24	17	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.0	3.8	8.4	15.6	25.7	26.8	28.6	27.5	19.8	12.7	8.5	4.4
		MCWB	0.7	2.3	3.2	8.3	15.2	17.4	20.3	19.3	15.1	10.2	7.3	3.6
	2%	DB	0.9	2.3	4.8	10.8	21.1	23.8	26.4	24.3	17.5	10.6	6.1	2.8
		MCWB	0.3	1.1	1.4	5.1	12.5	16.0	19.4	18.0	13.8	8.8	5.1	2.2
	5%	DB	0.2	1.3	3.2	8.8	18.5	21.7	24.7	22.3	16.0	9.3	4.8	1.7
		MCWB	-0.2	0.4	0.8	4.2	11.4	16.4	18.7	16.9	12.9	7.8	4.1	1.0
	10%	DB	-0.6	0.5	1.9	6.9	16.1	19.5	23.1	20.7	14.8	8.1	3.4	0.9
		MCWB	-1.0	-0.2	0.4	3.5	10.4	14.9	18.2	16.2	12.3	6.8	2.7	0.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.1	2.4	3.6	8.3	16.2	20.0	21.7	21.0	15.8	10.8	7.3	3.9
		MCDB	1.5	3.4	6.8	15.1	22.9	23.3	26.7	25.3	18.5	12.5	8.5	4.5
	2%	WB	0.4	1.2	2.1	6.1	13.8	18.0	20.5	18.9	14.5	9.3	5.4	2.3
		MCDB	0.8	2.0	4.0	9.6	19.1	22.7	24.8	22.8	16.6	10.2	6.1	2.7
	5%	WB	-0.2	0.5	1.2	4.8	12.2	16.5	19.7	17.6	13.6	7.9	4.1	1.2
		MCDB	0.2	1.2	2.7	7.8	17.4	20.9	23.7	21.0	15.4	8.8	4.8	1.6
	10%	WB	-1.1	-0.2	0.5	3.8	10.6	14.9	18.5	16.6	12.6	6.8	2.8	0.4
		MCDB	-0.5	0.6	1.9	6.5	15.4	18.7	22.3	20.2	14.4	8.1	3.4	0.9
Mean Daily Temperature Range	5% DB	MDBR	5.5	6.0	7.2	7.3	7.8	6.9	6.5	6.3	5.0	3.7	3.3	4.3
		MCDBR	3.5	4.6	6.4	9.6	11.9	9.9	9.1	8.7	7.0	5.2	4.7	3.5
		MCWBR	3.3	4.2	4.5	5.6	5.6	4.8	4.4	4.3	4.2	3.9	4.4	3.4
	5% WB	MCDBR	3.2	4.5	5.5	7.8	10.6	8.6	7.9	7.6	5.8	4.6	4.6	3.5
		MCWBR	3.0	4.2	4.4	5.1	6.2	4.9	4.3	4.5	4.0	4.1	4.4	3.5
Clear-Sky Solar Irradiance	taub	0.226	0.281	0.309	0.345	0.350	0.343	0.374	0.355	0.329	0.306	0.236	0.205	
	taud	1.987	2.097	2.158	2.288	2.447	2.496	2.425	2.455	2.513	2.394	2.069	1.956	
	Ebn at Noon	379	634	777	834	866	879	836	821	775	629	365	193	
	Edh at Noon	32	67	94	104	97	94	99	88	68	51	28	15	
All-Sky Solar Radiation	RadAvg	0.19	0.78	2.24	3.72	4.73	5.25	4.91	3.67	2.07	0.78	0.20	0.08	
	RadStd	0.02	0.16	0.25	0.34	0.51	0.39	0.53	0.37	0.22	0.10	0.04	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		+0.65	N/A	N/A	N/A	N/A	N/A	N/A	-227	N/A	N/A

Nomenclature: See separate page